

Code-Layout, Readability and Reusability

Date	1 July 2026
Team ID	
Project Name	Human Development Index (HDI) Prediction – A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the **Human Development Index (HDI) Prediction System** based on code organization, readability, and reusability. The project follows proper coding standards by using meaningful variable names, logical folder organization, reusable modules, and a structured Flask application. These practices make the project easy to understand, maintain, and extend in the future.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing used throughout the code	Yes	Code is properly formatted and easy to read
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for templates, static files, model, and application..
3	Meaningful Variable Names	Variables clearly describe their purpose	Yes	Variables such as life, expected, mean, gni, and prediction are descriptive.
4	Function / Method Names	Functions have meaningful names	Yes	Functions such as home() and predict() clearly indicate their purpose.
5	Code Comments	Comments explain important logic	Partial	Basic comments are included where required.
6	Modular Design	Code is divided into reusable modules	Yes	Machine learning model, Flask backend, HTML, and CSS are organized separately.
7	No Redundant Code	Duplicate or unused code removed	Yes	Only required code is retained.
8	Error Handling	Errors handled appropriately	Partial	Basic validation is implemented; advanced exception handling can be added later.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Linear Regression Model (HDI.pkl)	Python, Scikit-learn	Flask Prediction Module	High

2	Flask Backend (app.py)	Python, Flask	Handles user requests and predictions	High
3	HTML Templates (index.html, result.html)	HTML	User Interface	High
4	CSS Stylesheet (style.css)	CSS	Styling	High
5	Pickle Model Loader	Python (Pickle)	Loading trained model for prediction	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate modules.
Readability	5	Code is simple, clean, and uses meaningful names.
Reusability	5	Components such as the model, Flask backend, and CSS can be reused.
Documentation / Comments	4	Basic documentation provided; additional comments can improve clarity.
Overall Score	4.75/5	The project follows good coding practices and is easy to maintain and extend.